

NITESH KUMAR SINGH

Technical Business Analyst | Product Owner | Data Engineer

niteshkumarsingh.xlr@gmail.com

Certified SAFe 6 Agilist

+91 - 8208036878

PROFESSIONAL SUMMARY

Technical Business Analyst, Product Owner, and Data Engineer with 6+ years driving the discovery, documentation, and execution of business and functional requirements for global digital products. Skilled at running discovery sessions and requirement workshops, translating high-level business needs into clear specifications, and reconciling input from multiple stakeholders to align on a single direction – while also building the data pipelines and validating quality hands-on when needed. Partners closely with cross-functional, multi-disciplinary teams and often serves as the sole point of contact bridging siloed engineering groups across regions. Uses data analysis to measure success and diagnose root causes of complex problems. Experienced across IoT, e-commerce, and BI platforms, and comfortable running multiple projects simultaneously across India, Korea, and US time zones.

CORE COMPETENCIES

- **Product & Business Analysis:** Product Discovery & Requirement Definition, Business & Functional Requirements, Use Cases, User Stories & Acceptance Criteria, Workflow Analysis, Roadmapping & Prioritization, Stakeholder Engagement, Solution Evaluation, UAT Coordination, Requirement Lifecycle Management, Scrum Product Ownership
- **Tools:** Azure DevOps (ADO), JIRA, Microsoft Project (MPP) & Gantt, Advanced SQL, Power BI, Microsoft Fabric, Azure IoT Hub, Azure Synapse Analytics, Azure Stream Analytics, PostgreSQL, Python / PySpark, Database Management
- **Domain & Technical:** IoT & OTA Software Delivery, Device Twin, Cloud-to-Device Messaging, Telemetry Definition & Event Mapping, JSON Payload Design, Data Engineering (PySpark, Spark optimization, pipeline orchestration), ETL, Lakehouse, E-commerce & Consumer Experience
- **Wireframing & Modelling:** Axure, Balsamiq, MS Visio, Draw.io; Class, E-R, Flowchart, Activity, Data Flow, Sequence, Use Case diagrams, Wireframes, Prototypes
- **Testing & Quality:** Manual QA Testing, Functional Testing, UAT, V&V, Defect Triage, Release Readiness, Requirement Traceability

PROFESSIONAL EXPERIENCE

Visionet Systems Private Limited

2024 – Present

Technical Business Analyst (Product Owner / Data Engineer)

Web Portal Development – Remote Software Delivery Platform for Connected Devices

- Acted as **Technical Business Analyst and Product Owner** for a **global web portal** enabling remote software deployment, version control, and lifecycle monitoring for internet-connected (**IoT**) **medical devices** – driving discovery, documentation, and execution of functional and system requirements.
- Authored and maintained a **200+ page requirements specification** covering **portal workflows, screen-level behaviour, business rules, validations, edge cases, and acceptance criteria** – abstracting high-level business needs into digestible, testable detail for engineering.
- Defined requirements for **core product workflows** – device registration, deployment setup and approval, change-order cloning, batch selection, device eligibility and search, and end-to-end software-deployment tracking.
- Specified update-history tracking including **update lifecycle, device status, version history**, sourcing data from **Azure IoT Hub device twins persisted into SQL** for portal visibility and reporting.
- Specified surfacing of device diagnostic logs from blob storage directly in the portal, enabling self-service troubleshooting for end users.
- Led the **web portal redesign**, documenting UI behaviour, navigation flow, and screen-level changes to deliver an **improved, consumer-centric user experience**.
- Proactively evaluated **feasibility of new features and enhancements** with **cross-functional, multi-disciplinary teams** – product owners, developers, architects, ETL/BI teams, and subject-matter experts – reconciling conflicting inputs and mitigating technical risk to establish a **clear technical vision**.
- Operated as a **hybrid contributor** – spanning requirements, QA validation, hands-on debugging and reverse-engineering from source code, and data validation – filling gaps across the team to protect timelines and data integrity.
- Supported rollout by coordinating review and execution of **~800 test cases** within a compressed cycle; **owned data validation and QA during critical resource gaps** to protect integrity across environments.
- Triaged **complex production issues** with the development team – outlining **business impact and resolution options** to stakeholders – and drove **defect resolution, requirement clarification, UAT, and release readiness** to launch a **production-ready portal** supporting **regulatory compliance** and secure global deployment.
- Created and maintained **Verification & Validation (V&V) documentation** for the web portal, supporting **multiple regulatory testing cycles** and ensuring requirement traceability, test coverage, expected behaviour, and validation outcomes were properly documented.
- Supported **real-device testing for the web portal**, validating portal functionality, workflows, and device-data behaviour using actual connected devices to ensure **compliance readiness and production-level reliability**.
- Partnered with business teams on **user manuals and training documentation** so end users could efficiently adopt new features.
- Built and maintained the **project plan in Microsoft Project (MPP) with Gantt charts** to track effort, timelines, dependencies, and milestones across workstreams – giving stakeholders clear visibility into **schedule and delivery risk**.
- Managed all work items in **Azure DevOps** for transparency, audit readiness, and velocity tracking across releases while running multiple concurrent workstreams in a matrixed, **multi-time zone (India/Korea/US) environment**.
- Drove the **web portal base-language migration** from Korean to English to standardize terminology across the global application.
- Coordinated requirements for updating **UI labels, database values, stored procedures, and backend terminology**, and identified **migration-impact areas across front-end, database, and application layers**.
- Documented **migration requirements, validation points, and expected behaviour**, working with **cross-functional global teams** to ensure consistent English terminology.
- Conceptualized and defined the approach for enabling multiple language support on the web portal following the base-language migration, outlining how the platform could **support a scalable multilingual experience for global users**.

IoT Telemetry Architecture, Data Engineering & Cross-Team Integration

- Acted as the **single liaison across four independent product-family engineering teams** in Korea and the US team, becoming the **go-to device subject-matter expert** and the connective tissue between groups working in silos.
- Collaborated with **multiple cross-functional business teams** to identify, define, and document **use cases**, using them as the foundation for driving **new product development, solution design, and delivery alignment** across engineering, data, and BI teams.
- Architected the **telemetry data structure individually for each of four product families** – studying each product's service manual and system diagram to define how data looks in raw form and how it is transmitted, along with the **event mapping**.

- Defined **telemetry data models across multiple IoT product families**, documenting field names, business definitions, data types, formats, sources, sample values, validation rules, telemetry thresholds, acceptable ranges, and qualification rules to capture **device performance, usage, and session details**, and enable **validation, anomaly detection, and reliable dashboard interpretation**.
- Produced **telemetry event-mapping specifications** translating raw device events into meaningful **KPIs and dashboard insights** – defining event triggers, payload fields, timestamps, device identifiers, reporting metrics, and validating transformed outputs against source values and business rules.
- Designed **telemetry JSON payload structures** across product families and mapped attributes through **ETL transformation to BI reporting metrics**, supporting developers, ETL, and BI teams with sample payload references and **field-level consistency checks**.
- Defined the **telemetry data modelling and end-to-end data architecture** for raw telemetry ingestion from Azure IoT Hub, including how raw device data should be curated, enriched, sent to Cosmos DB, and further ingested into Delta tables in Microsoft Fabric through batch ingestion.
- Built the **synthetic telemetry generation system** as the **hands-on data engineer** – writing and optimizing the **PySpark code and orchestration** to produce realistic sample data conforming to defined business rules, unblocking BI dashboard development before live data existed.
- Defined and developed the **downstream fact tables and view table structures** required for **BI report consumption**, ensuring the curated telemetry data model supported **dashboard KPIs, filters, aggregations, and performance insights**.
- Drove **IoT BI reporting requirements** – KPIs, filters, dashboard views, and validation rules – and supported preliminary **Power BI dashboards on mock telemetry**, using data analysis to measure **product performance and surface operational insights**.
- Maintained **requirement traceability** across business needs, user stories, development tasks, test cases, telemetry outputs, and BI reporting.

Business Intelligence & Enterprise Reporting Platform

- Owned the **enterprise BI and reporting platform end to end** – source-side data ingestion and validation, ETL development, report development, and QA validation – delivering **10+** dashboards for **executive leadership and cross-functional teams** across Supply Chain, Finance, HR, Sales, and Quality.
- Gathered reporting requirements from diverse stakeholders and translated them into dashboards tracking key operational and business needs, then validated outputs across the full BI lifecycle from requirement understanding to final delivery.
- Built a **Sales Opportunity** dashboard tracking the complete lifecycle from **opportunity creation through deal closure to revenue generation**, giving leadership visibility into the sales pipeline and conversion.
- Developed **Global Orders and worldwide Installed Base dashboards to monitor order flow** and the deployed device footprint across regions.
- Delivered a dedicated **Quality dashboard** to track and categorize product complaints, enabling the Quality team to assess trends and prioritize business needs.
- Translated **complex SQL logic** into executive-ready visualizations and resolved critical reporting errors, including **multi-million-dollar** mismatches.
- Integrated **post-acquisition datasets**, strengthened data governance with **Row-Level Security (RLS)**, and standardized reporting logic using **reusable SQL views**.
- Wrote **validation scripts using advanced SQL** to support data validation, transformation validation, report validation, and business logic validation across the full journey from source systems to BI reporting.
- Partnered with **ETL/BI teams** to ensure end-to-end reporting accuracy and used Azure DevOps for documentation, QA, UAT, traceability, and post-deployment support.

Rego Digital Solutions Private Limited

2022 – 2024

Business Analyst

E-Commerce Shopping Website

- Developed a state-of-the-art online shopping website for seamless e-commerce transactions, providing a user-friendly interface with robust security measures, easy product navigation, search, and personalized recommendation **using Agile methodology**.
- Interacted with stakeholders and gathered requirements using various **elicitation techniques**.
- Created user stories with appropriate acceptance criteria with the assistance of the Product Owner. Added **user stories** into the **product backlog** using the **JIRA** tool.
- Prioritized and validated requirements using **MoSCoW** and **FURPS** techniques, and added **user stories** to the **sprint backlog** based on prioritization order.
- Collaborated with the Product Owner and Scrum Master for **BV** and **CP**, and assisted the Product Owner in the creation of the **DOR** and **DOD** checklist.
- Generated **Sprint and Product Burndown/Burnup** charts to track project progress. Participated in product planning and UAT to successfully deliver each sprint component.

Enhanced ED LMS Implementation

- Implemented a robust LMS that enhanced education delivery, content management, and user experience for students and educators by ensuring seamless integration with existing educational systems and tools, delivering a scalable and user-friendly platform **using Waterfall methodology**.
- Conducted **Enterprise Analysis** and assisted a senior BA in creating a Business Case Document, conducted Stakeholder Analysis, and prepared a RACI Matrix.
- Gathered requirements from business heads using elicitation techniques and created a **Business Requirements Document (BRD)**.
- Translated BRD into **Functional Requirement Document (FRD)**, collaborated with the technical team, and prepared **SRS Document**. Created and maintained **RTM** throughout the project.
- Assisted the Testing Team by preparing **test case scenarios** and ensured the **UAT** was successful.
- Created **UML diagrams** and **wireframes** to visually represent requirements using **MS Visio**, **Balsamiq**, and **Axure**.

IUVA Foods Private Limited

2020 – 2022

Data and Operation Head

- Normalized weekly unstructured CSV data into **3NF** using **PostgreSQL** and automated multi-source integration, improving database efficiency and consistency.
- Designed complex **SQL queries** and **batch scripts** for process improvements, reducing error rates and supporting strategic decisions.
- Enhanced pipeline efficiency with **automated SQL execution** and **built Power BI dashboards** for timely business intelligence.
- Improved onboarding completion from 35% to 60% and **optimized sales tracking**, driving a **20% increase** in outlet sales.
- Led geographic expansion to Bangalore, Mumbai, and Pune, established a vendor-management system, and supported a pre-revenue round **raising Rs. 1 Crore at a Rs. 5 Crore valuation**.
- Documented schemas, queries, and scripts, revised business strategies, and **reduced operational costs by 73%**.

EDUCATION

B.Tech – Mechanical Engineering

2014 – 2018

Manipal Institute of Technology, M.A.H.E

Senior Secondary (12th)

2011 – 2013

Navy Children School, Goa